



Rossmoyne Senior High School

Semester One Examination, 2022

Question/Answer booklet

MATHEMATICS METHODS UNIT 1

Section Two: Calculator-assumed

SOLUTIONS

WA student number: In figures

--	--	--	--	--	--	--	--

In words

--	--	--	--	--	--	--	--

Your name

--	--	--	--	--	--	--	--

Time allowed for this section

Reading time before commencing work: ten minutes

Working time: one hundred minutes

Number of additional
answer booklets used
(if applicable):

--

Materials required/recommended for this section

To be provided by the supervisor

This Question/Answer booklet

Formula sheet (retained from Section One)

To be provided by the candidate

Standard items: pens (blue/black preferred), pencils (including coloured), sharpener, correction fluid/tape, eraser, ruler, highlighters

Special items: drawing instruments, templates, notes on two unfolded sheets of A4 paper, and up to three calculators, which can include scientific, graphic and Computer Algebra System (CAS) calculators, are permitted in this ATAR course examination

Important note to candidates

No other items may be taken into the examination room. It is **your** responsibility to ensure that you do not have any unauthorised material. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

Structure of this paper

Section	Number of questions available	Number of questions to be answered	Working time (minutes)	Marks available	Percentage of examination
Section One: Calculator-free	7	7	50	54	35
Section Two: Calculator-assumed	12	12	100	96	65
Total					100

Instructions to candidates

1. The rules for the conduct of examinations are detailed in the school handbook. Sitting this examination implies that you agree to abide by these rules.
2. Write your answers in this Question/Answer booklet preferably using a blue/black pen. Do not use erasable or gel pens.
3. You must be careful to confine your answers to the specific question asked and to follow any instructions that are specific to a particular question.
4. Show all your working clearly. Your working should be in sufficient detail to allow your answers to be checked readily and for marks to be awarded for reasoning. Incorrect answers given without supporting reasoning cannot be allocated any marks. For any question or part question worth more than two marks, valid working or justification is required to receive full marks. If you repeat any question, ensure that you cancel the answer you do not wish to have marked.
5. It is recommended that you do not use pencil, except in diagrams.
6. Supplementary pages for planning/continuing your answers to questions are provided at the end of this Question/Answer booklet. If you use these pages to continue an answer, indicate at the original answer where the answer is continued, i.e. give the page number.
7. The Formula sheet is not to be handed in with your Question/Answer booklet.

Section Two: Calculator-assumed

65% (96 Marks)

This section has **twelve** questions. Answer **all** questions. Write your answers in the spaces provided.

Working time: 100 minutes.

Question 8

(6 marks)

Four points have coordinates $A(1, -1)$, $B(-14, 5)$, $C(s, t)$ and $D(3, 5)$.

- (i) If A is the midpoint of B and C , determine the value of the constant s and the value of the constant t . (2 marks)

Solution
$\frac{-14 + s}{2} = 1 \rightarrow s = 16$
$\frac{5 + t}{2} = -1 \rightarrow t = -7$
Specific behaviours
✓ value of s ✓ value of t (just comment if written as point and not values of constants)

- (ii) Determine the equation of the line that is perpendicular to AB and that passes through D in the form $ax + by + c = 0$, where a, b and c are integers and $a > 0$. (4 marks)

Solution
$m_{AB} = \frac{5 - (-1)}{-14 - 1} = -\frac{2}{5}, \quad m_{PERP} = -1 \div -\frac{2}{5} = \frac{5}{2}$
$y - 5 = \frac{5}{2}(x - 3)$
$2y - 10 = 5x - 15$
$5x - 2y - 5 = 0$
Specific behaviours
✓ slope of AB ✓ slope of perpendicular line ✓ correct equation of line, any form ✓ correct equation of line in required form

Question 9

(8 marks)

A calculator is used to randomly generate a whole number between 4 and 12 inclusive.

Event A occurs when the number is odd, event B occurs when the number is prime and event C occurs when the number is less than 9.

(a) Determine the ordered set of all possible outcomes for the event

(i) C

(1 mark)

Solution
$C = \{4, 5, 6, 7, 8\}$
Specific behaviours
✓ correct ordered list (<i>penalise lack of curly brackets once in whole question</i>)

(ii) $A \cup C$.

(1 mark)

Solution
$A \cup C = \{4, 5, 6, 7, 8, 9, 11\}$
Specific behaviours
✓ correct ordered list

(iii) $\bar{A} \cap \bar{B} \cap \bar{C}$.

(2 marks)

Solution
Numbers that are in none of the sets: $\bar{A} \cap \bar{B} \cap \bar{C} = \{10, 12\}$
Specific behaviours
✓ appropriate working ie list relevant sets ✓ correct ordered list

(b) Determine

(i) $P(B \cap C)$.

(2 marks)

Solution
Primes that are no more than 8: $B \cap C = \{5, 7\}$
$P(B \cap C) = \frac{n(B \cap C)}{n(U)} = \frac{2}{9}$
Specific behaviours
✓ identifies subset ✓ correct probability

(ii) $P(A \cap \overline{B \cup C})$.

(2 marks)

Solution
Odd numbers that are not prime and more than eight: $A \cap \overline{B \cup C} = \{9\}$
$P(A \cap \overline{B \cup C}) = \frac{1}{9}$
Specific behaviours
✓ identifies subset ✓ correct probability

See next page

Question 10

(8 marks)

Triangle ABC has sides $a = 17.2$ cm, $c = 21.6$ cm and an area of 171 cm².

- (a) Determine all possible values for the size of angle B in degrees.

(3 marks)

Solution
$171 = \frac{1}{2}(17.2)(21.6) \sin B$ $\sin B = 0.92054, \quad 0^\circ \leq B \leq 180^\circ$ $B = 67.0^\circ \text{ or } 113.0^\circ$
Specific behaviours
✓ substitutes into area formula correctly ✓ solves for one correct angle ✓ solves for both correct angles Allow any degree of accuracy

- (b) Determine, showing use of trigonometry, the largest possible perimeter of triangle ABC .

(3 marks)

Solution
$b^2 = 17.2^2 + 21.6^2 - 2(17.2)(21.6) \cos 113^\circ$ $b = 32.4$ $P = 17.2 + 21.6 + 32.4$ $= 71.2 \text{ cm}$
Specific behaviours
✓ shows use of trigonometry ✓ correct length of b ✓ correct perimeter with units If used 67° then $b=21.7$ and P 60.5 cm max 2/3 marks

- (c) Determine, showing use of trigonometry, the size of angle A when triangle ABC has the largest possible perimeter.

(2 marks)

Solution
$\frac{\sin A}{17.2} = \frac{\sin 113^\circ}{32.4}$ $A = 29.2^\circ$
Specific behaviours
✓ shows use of trigonometry ✓ solves for correct angle (no marks for considering ambiguous case as already awarded in (a)) If used 67° then $A=47^\circ$ allow fullmarks follow through

Question 11**(10 marks)**

The time of sunrise, t hours after midnight, on the n^{th} day of the year in Esperance is closely modelled by

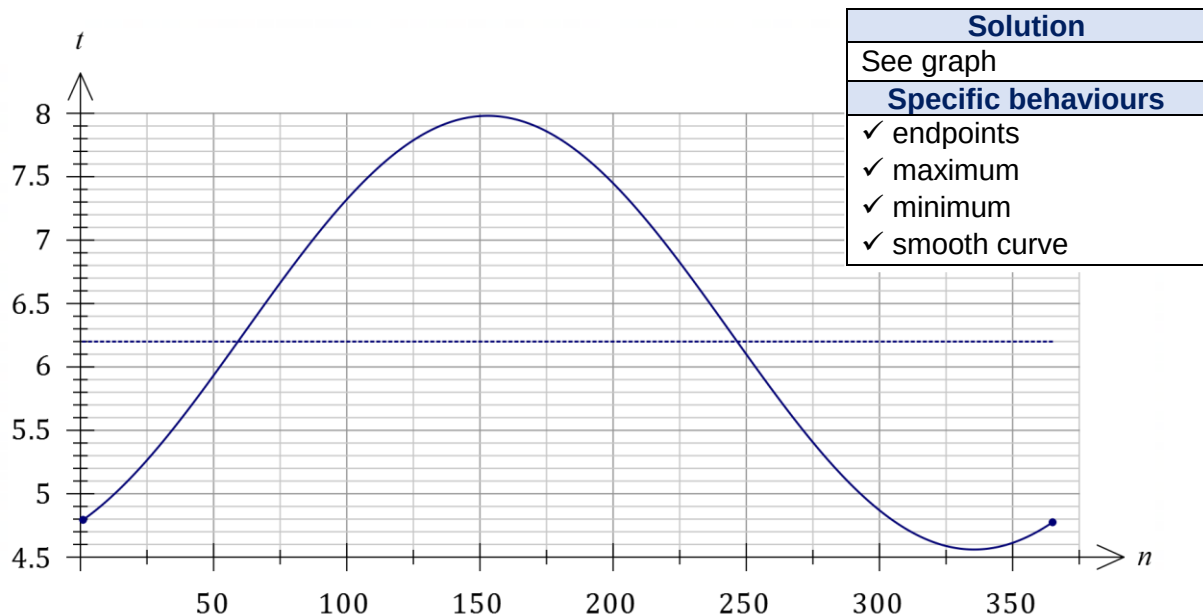
$$t = 6.27 - 1.71 \sin(0.0172n + 2.083).$$

- (a) Use the model to calculate, in hours and minutes, the time of sunrise on 5 February rounding to the nearest minute.

(2 marks)

Solution	
$n = 31 + 5 = 36, \quad t(36) = 5.543 = 5:33 \text{ am}$	
Specific behaviours	
✓ correct value of n ✓ correct time in hours and minutes rounded correctly	

- (b) Graph t as a function of n on the axes below for $0 \leq n \leq 365$.

(4 marks)

- (c) Jill lives in Esperance and is woken by her alarm clock at 6:12 am every day. How many days will she have woken up after sunrise in a year of 365 days? **(4 marks)**

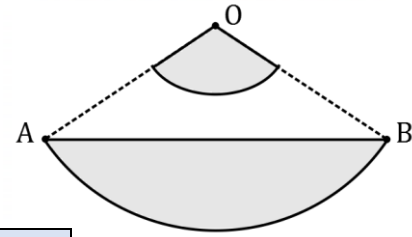
Solution	
6:12 am $\rightarrow$ 6.2 h, $t = 6.2$ when $n = 59.17, n = 246.58$ (see dotted line on graph). $6.12 = 6.27 - 1.71 \sin(0.0172n + 2.083)$	
$t < 6.2$ when $n \leq 59, n \geq 247$. Hence awake days 1 through 59 and 247 through 365, which is 178 days.	
Specific behaviours	
✓ indicates 6:12 am $\rightarrow$ 6.2 h, ✓ solves for $t = 6.2$ ✓ indicates correct intervals of days ✓ correct number of days, allow 177 if used 6:12 then 56 to 250 range giving $56 + 116 = 172$ so give $\frac{3}{4}$ marks	

Question 12

(9 marks)

The shaded regions shown in the diagram form a logo that is to be cut from a plastic sheet.

The boundaries of the shaded regions are parts of triangle AOB , where $\angle AOB = 110^\circ$, and arcs of two concentric circles with centre O and radii of 35 cm and 91 cm.



- (a) Express 110° exactly in radians.

Solution	
$110^\circ = \frac{11\pi}{18}$	
Specific behaviours	
✓ correct conversion	

(1 mark)

- (b) Determine the total length of the two curved and three straight cuts that must be made by the cutting machine, giving your answer to the nearest centimetre. (4 marks)

Solution	
Length of arcs ($L = r\theta$):	
$L_1 = 35 \times \frac{11\pi}{18} = 67.2, \quad L_2 = 91 \times \frac{11\pi}{18} = 174.7$	
Length of AB :	
$AB^2 = 91^2 + 91^2 - 2(91)(91)\cos\left(\frac{11\pi}{18}\right)$	
$AB = 149.1$	
Total lengths of cuts: $L = L_1 + L_2 + AB + 2(35) = 461$ cm.	
Specific behaviours	
✓ calculates one correct arc length	
✓ indicates use of trigonometry	
✓ correct length of AB	
✓ correct total length	

- (c) Determine the area of logo, giving your answer to the nearest square centimetre.

(4 marks)

Solution	
Area of sector:	
$A_1 = \frac{1}{2}(35)^2 \times \frac{11\pi}{18} = 1175.9$	
Area of segment:	
$A_2 = \frac{1}{2}(91)^2 \left(\frac{11\pi}{18} - \sin\left(\frac{11\pi}{18}\right) \right) = 4058.4$	
Total area: $A = A_1 + A_2 = 5234$ cm ² .	
Specific behaviours	
✓ correct area of sector	
✓ indicates correct use of segment area formula	
✓ correct area of segment	
✓ adds segment and sector to obtain correct total area	

Question 13**(7 marks)**

In a random sample of 315 cats, it was found that 107 of the cats were males with green-coloured eyes, and 92 of the 148 females had green-coloured eyes.

(a) Complete all the regions in the Venn diagram to represent this situation.

(2 marks)

Gender	Eye Colour		
	Green	Not Green	Total
Male	107	60	167
Female	92	56	148
Total	199	116	315

Solution
Specific behaviours
✓ 107 and 92 correct ✓ all correct

(b) Determine the probability that a randomly selected cat from the sample

(i) is female.

(1 mark)

Solution
$P(F) = \frac{148}{315} \approx 0.4698$
Specific behaviours
✓ correct probability

(ii) is female or does not have green-coloured eyes.

(2 marks)

Solution
$ \begin{aligned} P(F \cup \bar{G}) &= P(F) + P(\bar{F} \cap \bar{G}) \\ &= \frac{148 + 60}{315} \\ &= \frac{208}{315} \approx 0.6603 \end{aligned} $
Specific behaviours
✓ correct numerator ✓ correct probability

(iii) is female given that it has green-coloured eyes.

(2 marks)

Solution
$ \begin{aligned} P(F G) &= \frac{P(F \cap G)}{P(G)} \\ &= \frac{92}{92 + 107} = \frac{92}{199} \approx 0.4623 \end{aligned} $
Specific behaviours
✓ correct denominator 199 ✓ correct numerator 92

Question 14

(9 marks)

A function defined by $f(x) = \frac{a}{x+b}$, where a and b are constants, passes through the points $(-8, 0.2)$ and $(2, -0.6)$.

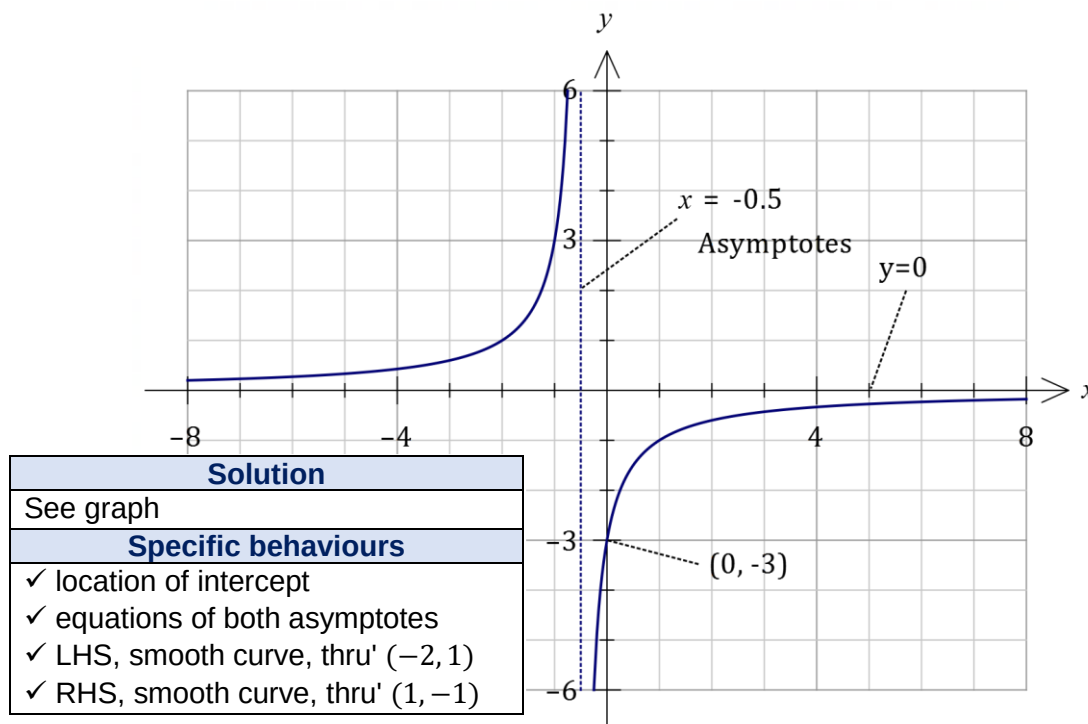
- (a) Determine the value of a and the value of b .

(3 marks)

Solution
$0.2 = \frac{a}{-8+b}, \quad -0.6 = \frac{a}{2+b}$ Solve simultaneously using CAS: $a = -1.5, b = 0.5$
Specific behaviours
✓ uses points to form two equations ✓ value of a ✓ value of b

- (b) Draw the graph of $y = f(x)$ on the axes below, clearly indicating the coordinates of all axes intercepts and equations of any asymptotes.

(4 marks)



- (c) State the equations of all asymptotes of the graph of $y = f(2x) - 3$.

(2 marks)

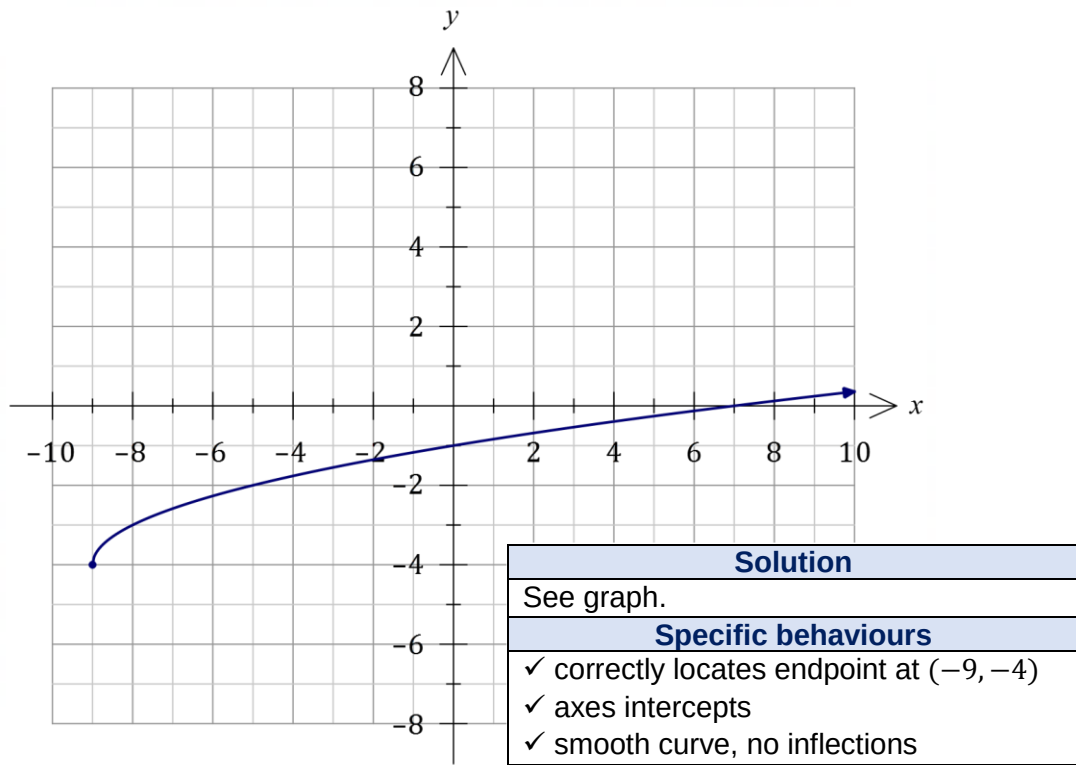
Solution
$y = 0 - 3 \rightarrow y = -3, \quad x = -\frac{1}{2} \div 2 \rightarrow x = -\frac{1}{4}$
Specific behaviours
✓ equations of horizontal asymptote ✓ equations of vertical asymptote

Question 15

(7 marks)

Let $f(x) = \sqrt{x+9} - 4$.(a) Sketch the graph of $y = f(x)$ on the axes below.

(3 marks)



(b) State the domain and range of

(i) $f(3x) - 3$.

(2 marks)

Solution
Horizontal dilation scale factor $\frac{1}{3}$: $D_f = \{x: x \in \mathbb{R}, x \geq -3\}$ Vertical translation 3 units downwards: $R_f = \{y: y \in \mathbb{R}, y \geq -7\}$
Specific behaviours
✓ correct domain ✓ correct range just inequalities is OK but -1 once if not referred as domain or range

(ii) $4f(x-4)$.

(2 marks)

Solution
Horizontal translation 4 units right: $D_f = \{x: x \in \mathbb{R}, x \geq -5\}$ Vertical dilation scale factor 4: $R_f = \{y: y \in \mathbb{R}, y \geq -16\}$
Specific behaviours
✓ correct domain ✓ correct range

Question 16

(8 marks)

- (a) The graph of the quadratic function $y = f(x)$ has roots at $x = -7$ and $x = 3$ and the range of $f(x)$ is $y \leq 50$. Use an algebraic method to determine $f(0)$. (4 marks)

Solution
Axis of symmetry is midway between roots $(-7 + 3) \div 2 = -2$ and so turning point at $(-2, 50)$. $f(x) = a(x + 2)^2 + 50$ Using a root, $f(3) = 0 = a(3 + 2)^2 + 50 \rightarrow a = -2$. $f(x) = -2(3 + 2)^2 + 50$ Hence $f(0) = -2(0 + 2)^2 + 50 = 42$.
Specific behaviours
✓ uses range and symmetry to identify turning point ✓ writes equation of quadratic with constant a ✓ evaluates constant a ✓ determines $f(0) = 42$

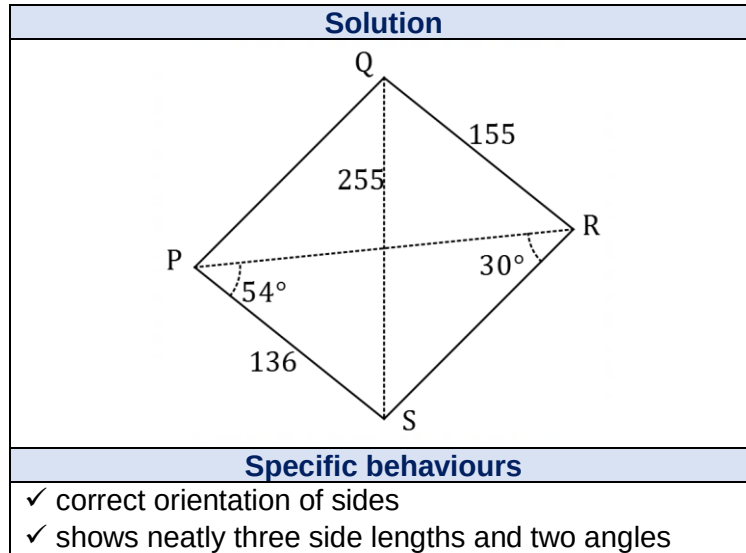
- (b) The area of square B is 655.5 cm^2 more than six times the area of square A , and the difference in the perimeters of the two squares is 98 cm . Determine the least possible area of square A , the smaller of the squares. (4 marks)

Solution
Side of B must be $98 \div 4 = 24.5 \text{ cm}$ longer than side of A. If side of A is x, then side of B is $x + 24.5$. Hence $(x + 24.5)^2 = 6x^2 + 655.5$ $x = \frac{13}{10} = 1.3, \quad x = \frac{17}{2} = 8.5$ Least area when $x = 1.3$: $A_A = 1.3^2 = \frac{169}{100} = 1.69 \text{ cm}^2$
Specific behaviours
✓ indicates difference in the side lengths is 24.5 ✓ forms correct quadratic equation ✓ solves quadratic equation ✓ states correct least area for simultaneous eqns: 1 mark each eqn, 1 mark for solve for x, 1 mark correct area

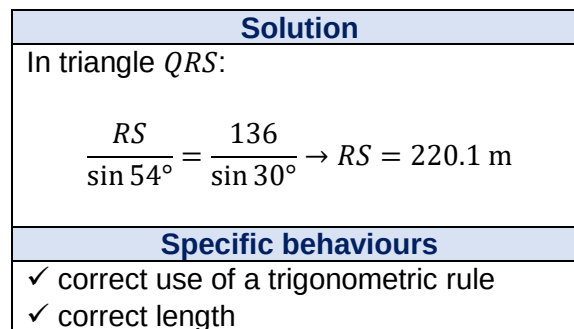
Question 17**(7 marks)**

A block of land $PQRS$ is bounded by four straight sides, with P the most westerly corner. The lengths of sides PS and QR are 136 m and 155 m respectively and Q lies 255 m due north of S . Diagonal PR makes an angle of 54° with PS and 30° with RS .

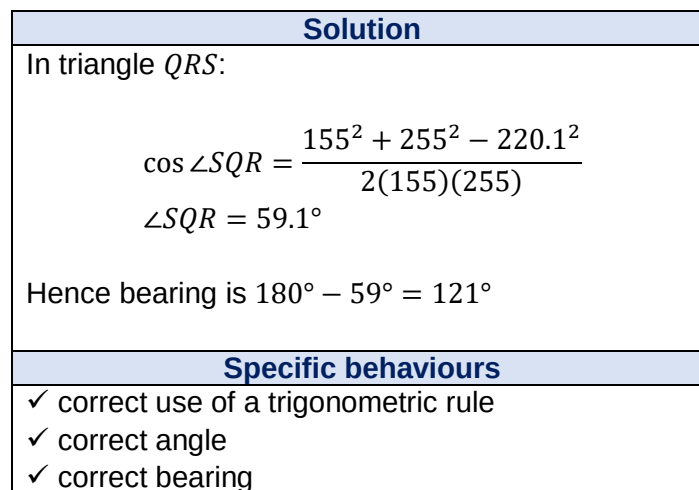
- (a) Sketch a diagram of the block, showing all the above information.

(2 marks)

- (b) Determine the length of side RS .

(2 marks)

- (c) Determine the bearing of R from Q .

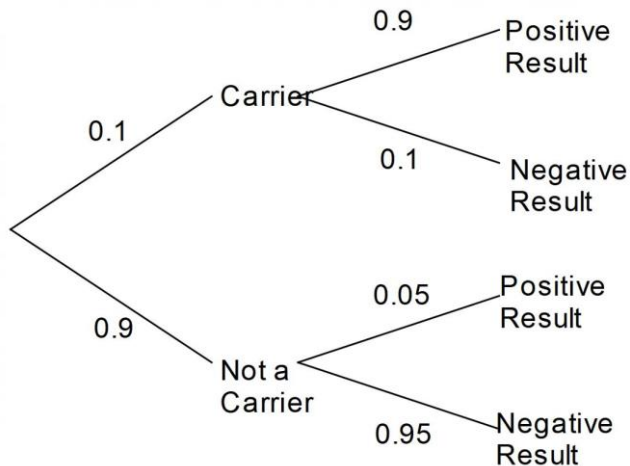
(3 marks)

Question 18

(7 marks)

Researchers found that 10% of rock lobsters were carriers of a newly detected shell disease. The researchers trialed a test to detect whether or not a lobster was a carrier of the disease. It was found that the test returned a positive result in 90% of the lobsters who were carriers of the disease but unfortunately incorrectly returned a negative result in the other 10%. When the test was administered with lobsters that were known not to be carriers of the disease, 95% came back as a negative result but for the other 5% an incorrect positive result was returned.

- (a) Complete the tree diagram below showing all probabilities and outcomes. (3 marks)



Solution
Specific behaviours
✓ correct outcomes ✓ correct 0.9 for not a carrier ✓ all correct probabilities

- (b) What is the probability that the test will return a correct result? (2 marks)

Solution
$P(\text{Correct}) = 0.1 \times 0.9 + 0.9 \times 0.95 = 0.945$
Specific behaviours
✓ correct prob for carrier and pos 0.1×0.9 ✓ correct answer

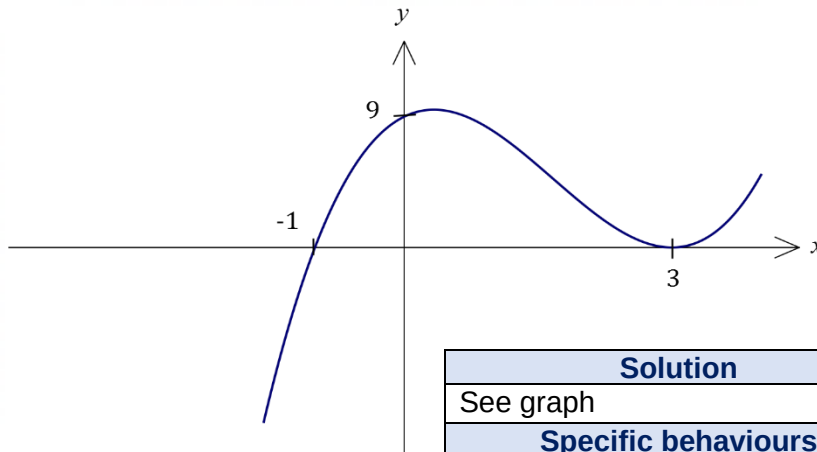
- (c) If a lobster returns a positive result, what is the probability that they are a carrier? (2 marks)

Solution
$P(\text{carr/pos}) = \frac{0.1 \times 0.9}{0.1 \times 0.9 + 0.9 \times 0.05} = \frac{0.09}{0.135} = \frac{2}{3}$
Specific behaviours
✓ correct denominator ✓ correct answer

Question 19**(10 marks)**

The graph of the cubic polynomial $y = f(x)$ passes through the points $(-1, 0)$, $(0, 9)$ and has a minimum turning point at $(3, 0)$.

(a) Use only the above information to sketch the graph of $y = f(x)$ on the axes below.

(3 marks)

Solution
See graph
Specific behaviours
✓ smooth cubic curve ✓ graph tangent to x-axis at $x=3$ ✓ locates both other intercepts

Let $f(x) = x^3 + bx^2 + cx + d$, where b, c and d are constants.

(b) Determine the value of each of the constants b, c and d .

(3 marks)

Solution
Factored form of cubic is $f(x) = (x - 3)^2(x + 1)$
Hence $f(x) = x^3 - 5x^2 + 3x + 9$
And so $b = -5$, $c = 3$, $d = 9$
Specific behaviours
✓ correct factored form of cubic ✓ expands cubic ✓ correct value for each constant

- (c) Another cubic polynomial is defined by $g(x) = x^3 - 6x^2 + ax + 5$. Determine the value(s) of the constant a so that the graphs of $y = f(x)$ and $y = g(x)$ do not intersect. (4 marks)

Solution
For intersection require $f = g \rightarrow x^3 - 5x^2 + 3x + 9 = x^3 - 6x^2 + ax + 5$. Hence $x^2 + (3 - a)x + 4 = 0$. For no intersection, this quadratic must have no solution and so discriminant, $\Delta = (3 - a)^2 - 4(1)(4)$, must be less than zero. Hence $(3 - a)^2 < 16$ and so $-1 < a < 7$.
Specific behaviours
✓ equates cubic equations and obtains quadratic ✓ forms discriminant ✓ recognises discriminant must be less than zero ✓ correct range of values for a If use simultaneous eqns: 1 mark 1 eqn, 1 mark 3 correct eqns, 1 mark correct soln

Supplementary page

Question number: _____

